

Concept 3 | Polynomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Remainder

$$P(a) = R \text{ when divisor} = x - a$$

Factor

$$P(a) = 0 \Rightarrow (x - a) \mid P(x)$$

Roots

$$P(x) = a \prod (x - r_i)$$

Worked Solutions

Polynomial Theorems

Q001 **Written****Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$.****Solution**

1. $U_{\text{sex}} = -2$
2. $P(-2) = 2$
3. Therefore the final answer is 2.

Final answer

2

Q002 **Written****Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$.****Solution**

1. $U_{\text{sex}} = 1$
2. $P(1) = 6$
3. Therefore the final answer is 6.

Final answer

6

Worked Solutions

Polynomial Theorems

Q003 MCQ

Find the remainder when $x^3 + 2x^2 - 3x - 4$ is divided by $x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 2.

Correct option: C**Final answer**

2

Q004 MCQ

Find the remainder when $3x^3 + 2x^2 - 5x + 6$ is divided by $x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 6.

Correct option: C**Final answer**

6

Worked Solutions

Polynomial Theorems

Q005 **Written****Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$.****Solution**

1. $Use x = 2/3$
2. $Evaluate P(2/3) = 8$
3. Therefore the final answer is 8.

Final answer

8

Q006 **Written****Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$.****Solution**

1. $Use x = 1/2$
2. $P(1/2) = -1/8$
3. Therefore the final answer is $-\frac{1}{8}$.

Final answer $-\frac{1}{8}$

Worked Solutions

Polynomial Theorems

Q007 **Written****Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$.****Solution**

1. Use $x = 2/3$
2. $P(2/3) = 0$
3. Therefore the final answer is 0.

Final answer

0

Q008 **Written****Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$.****Solution**

1. Use $x = -1/2$
2. $P(-1/2) = 4$
3. Therefore the final answer is 4.

Final answer

4

Worked Solutions

Polynomial Theorems

Q009 MCQ

Find the remainder when $3x^4 - 8x^3 - 5x^2 + 12$ is divided by $3x - 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 8.

Correct option: D**Final answer**

8

Q010 MCQ

Find the remainder when $x^3 - 5x^2 + 4x - 1$ is divided by $2x - 1$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $-\frac{1}{8}$.

Correct option: A**Final answer** $-\frac{1}{8}$

Worked Solutions

Polynomial Theorems

Q011 MCQ

Find the remainder when $6x^3 - x^2 + x - 2$ is divided by $3x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 0.

Correct option: D**Final answer**

0

Q012 MCQ

Find the remainder when $4x^3 - 2x^2 - 4x + 3$ is divided by $2x + 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 4.

Correct option: A**Final answer**

4

Worked Solutions

Polynomial Theorems

Q013 Written

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a .

Solution

1. $P(-3) = 4$
2. $9a - 35 = 4$, hence $a = 13/3$
3. Therefore the final answer is $a = 13/3$.

Final answer

$$a = 13/3$$

Q014 Written

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a .

Solution

1. $P(3) = 1$
2. $9a + 37 = 1$, hence $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$$a = -4$$

Worked Solutions

Polynomial Theorems

Q015 Written

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2 , find a .

Solution

1. $P(-1) = -2$
2. $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$$a = -4$$

Q016 Written

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8 , find a .

Solution

1. $P(-3) = 8$
2. $a = -16$
3. Therefore the final answer is $a = -16$.

Final answer

$$a = -16$$

Worked Solutions

Polynomial Theorems

Q017 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x + 3$ is 4, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = 13/3$.

Correct option: B**Final answer**

$$a = 13/3$$

Q018 MCQ

If the remainder when $x^3 + ax^2 + 3x + 1$ is divided by $x - 3$ is 1, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -4$.

Correct option: B**Final answer**

$$a = -4$$

Worked Solutions

Polynomial Theorems

Q019 MCQ

If the remainder when $x^3 - x^2 + ax - 4$ is divided by $x + 1$ is -2 , find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -4$.

Correct option: C**Final answer**

$$a = -4$$

Q020 MCQ

If the remainder when $x^3 + 4x^2 - 5x + a$ is divided by $x + 3$ is 8 , find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -16$.

Correct option: A**Final answer**

$$a = -16$$

Worked Solutions

Polynomial Theorems

Q021 Written

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a .

Solution

1. $P(2) = 0$
2. $a = -4$
3. Therefore the final answer is $a = -4$.

Final answer

$$a = -4$$

Q022 MCQ

If $x^3 - 3x^2 + 4x + a$ is divisible by $x - 2$, find a Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -4$.

Correct option: A

Final answer

$$a = -4$$

Worked Solutions

Polynomial Theorems

Q023 Written

A polynomial $P(x)$ leaves remainders 3 on division by $x - 1$ and -5 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

Solution

1. Let $R = ax + b$
2. $a + b = 3$ and $-3a + b = -5$
3. $R = 2x + 1$
4. Therefore the final answer is $2x + 1$.

Final answer

$$2x + 1$$

Q024 Written

A polynomial $P(x)$ leaves remainders 3 on division by $x - 2$ and -7 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

Solution

1. Let $R = ax + b$
2. $2a + b = 3$ and $-3a + b = -7$
3. $R = -10x + 13$
4. Therefore the final answer is $-10x + 13$.

Final answer

$$-10x + 13$$

Worked Solutions

Polynomial Theorems

Q025 **Written**

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$.

Solution

1. Let $R = ax + b$
2. $a + b = 5$ and $-2a + b = -1$
3. $R = 2x + 3$
4. Therefore the final answer is $2x + 3$.

Final answer

$$2x + 3$$

Q026 **Written**

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$.

Solution

1. Let $R = ax + b$
2. $2a + b = 4$ and $-3a + b = -11$
3. $R = -15x + 19$
4. Therefore the final answer is $-15x + 19$.

Final answer

$$-15x + 19$$

Worked Solutions

Polynomial Theorems

Q027 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x - 1$ and -5 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $2x + 1$.

Correct option: A

Final answer

$$2x + 1$$

Q028 MCQ

A polynomial $P(x)$ leaves remainders 3 on division by $x - 2$ and -7 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $-10x + 13$.

Correct option: A

Final answer

$$-10x + 13$$

Worked Solutions

Polynomial Theorems

Q029 MCQ

A polynomial $P(x)$ leaves remainders 5 on division by $x - 1$ and -1 on division by $x + 2$. Find the remainder on division by $(x - 1)(x + 2)$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $2x + 3$.

Correct option: A

Final answer

$$2x + 3$$

Q030 MCQ

A polynomial $P(x)$ leaves remainders 4 on division by $x - 2$ and -11 on division by $x + 3$. Find the remainder on division by $x^2 + x - 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $-15x + 19$.

Correct option: A

Final answer

$$-15x + 19$$

Worked Solutions

Polynomial Theorems

Q031 Written

A polynomial $P(x)$ leaves remainder $7x + 6$ on division by $x^2 - x - 6$, and $4x - 3$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 + x - 2$.

Solution

1. Let $R = ax + b$
2. Use shared roots -2 and 1 : $-2a + b = -8$ and $a + b = 1$
3. $R = 3x - 2$
4. Therefore the final answer is $3x - 2$.

Final answer

$$3x - 2$$

Q032 Written

A polynomial $P(x)$ leaves remainder $3x + 2$ on division by $x^2 - x - 2$, and $7x - 6$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 - 1$.

Solution

1. Let $R = ax + b$
2. At -1 and 1 : $-a + b = -1$ and $a + b = 1$
3. $R = x$
4. Therefore the final answer is x .

Final answer

$$x$$

Worked Solutions

Polynomial Theorems

Q033 Written

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$.

Solution

1. Use shared roots -1 and 2
2. $-a + b = -4$ and $2a + b = 5$, so $R = 3x - 1$
3. Therefore the final answer is $3x - 1$.

Final answer

$$3x - 1$$

Q034 Written

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$.

Solution

1. Use shared roots 2 and 3
2. $2a + b = 2$ and $3a + b = 9$, so $R = 7x - 12$
3. Therefore the final answer is $7x - 12$.

Final answer

$$7x - 12$$

Worked Solutions

Polynomial Theorems

Q035 MCQ

A polynomial $P(x)$ leaves remainder $7x + 6$ on division by $x^2 - x - 6$, and $4x - 3$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 + x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $3x - 2$.

Correct option: B

Final answer $3x - 2$

Q036 MCQ

A polynomial $P(x)$ leaves remainder $3x + 2$ on division by $x^2 - x - 2$, and $7x - 6$ on division by $x^2 + 2x - 3$. Find the remainder on division by $x^2 - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is x .

Correct option: B

Final answer x

Worked Solutions

Polynomial Theorems

Q037 MCQ

A polynomial $P(x)$ leaves remainder $2x - 2$ on division by $x^2 - 2x - 3$, and $-x + 7$ on division by $x^2 - 4$. Find the remainder on division by $x^2 - x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $3x - 1$.

Correct option: A

Final answer

$$3x - 1$$

Q038 MCQ

A polynomial $P(x)$ leaves remainder $-x + 4$ on division by $x^2 - 3x + 2$, and $3x$ on division by $x^2 - 4x + 3$. Find the remainder on division by $x^2 - 5x + 6$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $7x - 12$.

Correct option: D

Final answer

$$7x - 12$$

Worked Solutions

Polynomial Theorems

Q039 Written

A polynomial $P(x)$ leaves remainder $x - 5$ on division by $(x + 1)^2$, and 6 on division by $x - 2$. Find the remainder on division by $(x + 1)^2(x - 2)$.

Solution

1. Let $R = ax^2 + bx + c$
2. $R(-1) = -6, R'(-1) = 1, R(2) = 6$
3. Solve to get $R = x^2 + 3x - 4$
4. Therefore the final answer is $x^2 + 3x - 4$.

Final answer

$$x^2 + 3x - 4$$

Q040 MCQ

A polynomial $P(x)$ leaves remainder $x - 5$ on division by $(x + 1)^2$, and 6 on division by $x - 2$. Find the remainder on division by $(x + 1)^2(x - 2)$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $x^2 + 3x - 4$.

Correct option: A

Final answer

$$x^2 + 3x - 4$$

Worked Solutions

Polynomial Theorems

Q041 Written

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$.

Solution

1. Use -2 and coefficients 1,0,5,11
2. Bottom row 1,-2,9,-7
3. Therefore the final answer is $Q = x^2 - 2x + 9, R = -7$.

Final answer

$$Q = x^2 - 2x + 9, R = -7$$

Q042 MCQ

Using synthetic division, find the quotient and remainder when $x^3 + 5x + 11$ is divided by $x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = x^2 - 2x + 9, R = -7$.

Correct option: A

Final answer

$$Q = x^2 - 2x + 9, R = -7$$

Worked Solutions

Polynomial Theorems

Q043 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 1,0,5,-6
2. Bottom row 1,1,6,0
3. Therefore the final answer is $Q = x^2 + x + 6, R = 0$.

Final answer

$$Q = x^2 + x + 6, R = 0$$

Q044 Written

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$.

Solution

1. Use 3 and coefficients 3,-5,-3,-1
2. Bottom row 3,4,9,26
3. Therefore the final answer is $Q = 3x^2 + 4x + 9, R = 26$.

Final answer

$$Q = 3x^2 + 4x + 9, R = 26$$

Worked Solutions

Polynomial Theorems

Q045 Written

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$.

Solution

1. Use 2 and coefficients 1,0,-5,6
2. Bottom row 1,2,-1,4
3. Therefore the final answer is $Q = x^2 + 2x - 1, R = 4$.

Final answer

$$Q = x^2 + 2x - 1, R = 4$$

Q046 Written

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 1,2,-1,-3
2. Bottom row 1,3,2,-1
3. Therefore the final answer is $Q = x^2 + 3x + 2, R = -1$.

Final answer

$$Q = x^2 + 3x + 2, R = -1$$

Worked Solutions

Polynomial Theorems

Q047 **Written**

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$.

Solution

1. Use 1 and coefficients 4,-2,2
2. Bottom row 4,2,4
3. Therefore the final answer is $Q = 4x + 2, R = 4$.

Final answer

$$Q = 4x + 2, R = 4$$

Q048 **MCQ**

Using synthetic division, find the quotient and remainder when $A = x^3 + 5x - 6$ is divided by $B = x - 1$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = x^2 + x + 6, R = 0$.

Correct option: B

Final answer

$$Q = x^2 + x + 6, R = 0$$

Worked Solutions

Polynomial Theorems

Q049 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^3 - 5x^2 - 3x - 1$ is divided by $B = x - 3$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = 3x^2 + 4x + 9, R = 26$.

Correct option: D**Final answer**

$$Q = 3x^2 + 4x + 9, R = 26$$

Q050 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 - 5x + 6$ is divided by $B = x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = x^2 + 2x - 1, R = 4$.

Correct option: D**Final answer**

$$Q = x^2 + 2x - 1, R = 4$$

Worked Solutions

Polynomial Theorems

Q051 MCQ

Using synthetic division, find the quotient and remainder when $A = x^3 + 2x^2 - x - 3$ is divided by $B = x - 1$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = x^2 + 3x + 2, R = -1$.

Correct option: A

Final answer

$$Q = x^2 + 3x + 2, R = -1$$

Q052 MCQ

Using synthetic division, find the quotient and remainder when $A = 4x^2 - 2x + 2$ is divided by $B = x - 1$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = 4x + 2, R = 4$.

Correct option: B

Final answer

$$Q = 4x + 2, R = 4$$

Worked Solutions

Polynomial Theorems

Q053 Written

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$.

Solution

1. Use -2 and coefficients 3,5,0,-2,-12
2. Bottom row 3,-1,2,-6,0
3. Therefore the final answer is $Q = 3x^3 - x^2 + 2x - 6, R = 0$.

Final answer

$$Q = 3x^3 - x^2 + 2x - 6, R = 0$$

Q054 MCQ

Using synthetic division, find the quotient and remainder when $A = 3x^4 + 5x^3 - 2x - 12$ is divided by $B = x + 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $Q = 3x^3 - x^2 + 2x - 6, R = 0$.

Correct option: D

Final answer

$$Q = 3x^3 - x^2 + 2x - 6, R = 0$$

Worked Solutions

Polynomial Theorems

Q055 Written

Factorise $x^3 + 2x^2 - x - 2$.**Solution**

1. $P(1) = 0$; *divide by* $x - 1$
2. *Quotient* $x^2 + 3x + 2 = (x + 1)(x + 2)$
3. Therefore the final answer is $(x - 1)(x + 1)(x + 2)$.

Final answer

$$(x - 1)(x + 1)(x + 2)$$

Q056 Written

Factorise $x^3 - 2x^2 - 5x + 6$.**Solution**

1. $P(1) = 0$; *quotient* $x^2 - x - 6 = (x - 3)(x + 2)$
2. Completed factorisation
3. Therefore the final answer is $(x - 1)(x - 3)(x + 2)$.

Final answer

$$(x - 1)(x - 3)(x + 2)$$

Worked Solutions

Polynomial Theorems

Q057 Written

Factorise $x^3 - x^2 - 16x - 20$.**Solution**

1. $P(-2) = 0$; divide by $x + 2$ twice
2. Remaining $x - 5$
3. Therefore the final answer is $(x + 2)^2(x - 5)$.

Final answer

$$(x + 2)^2(x - 5)$$

Q058 Written

Factorise $2x^3 - x^2 - 5x - 2$.**Solution**

1. $P(2) = 0$; quotient $2x^2 + 3x + 1$
2. Therefore the final answer is $(x - 2)(2x + 1)(x + 1)$.

Final answer

$$(x - 2)(2x + 1)(x + 1)$$

Worked Solutions

Polynomial Theorems

Q059 Written

Factorise $x^3 - 7x - 6$.**Solution**

1. Test divisor 3 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x - 3)(x + 1)(x + 2)$.

Final answer

$$(x - 3)(x + 1)(x + 2)$$

Q060 Written

Factorise $x^3 + 4x^2 + x - 6$.**Solution**

1. Test divisor 1 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x - 1)(x + 2)(x + 3)$.

Final answer

$$(x - 1)(x + 2)(x + 3)$$

Worked Solutions

Polynomial Theorems

Q061 Written

Factorise $x^3 + 6x^2 + 11x + 6$.**Solution**

1. Test divisor -1 and divide
2. Remaining linear factors
3. Therefore the final answer is $(x + 1)(x + 2)(x + 3)$.

Final answer

$$(x + 1)(x + 2)(x + 3)$$

Q062 Written

Factorise $x^3 - 3x + 2$.**Solution**

1. Test divisor 1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 1)^2(x + 2)$.

Final answer

$$(x - 1)^2(x + 2)$$

Worked Solutions

Polynomial Theorems

Q063 Written

Factorise $x^3 + x^2 - 8x - 12$.**Solution**

1. Test divisor -2 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 3)(x + 2)^2$.

Final answer

$$(x - 3)(x + 2)^2$$

Q064 Written

Factorise $x^3 - 3x^2 - 9x - 5$.**Solution**

1. Test divisor -1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x - 5)(x + 1)^2$.

Final answer

$$(x - 5)(x + 1)^2$$

Worked Solutions

Polynomial Theorems

Q065 Written

Factorise $2x^3 - 3x^2 - 3x + 2$.**Solution**

1. Test divisor 2 and divide
2. *Factorquotient* $2x^2 + x - 1$
3. Therefore the final answer is $(x - 2)(x + 1)(2x - 1)$.

Final answer

$$(x - 2)(x + 1)(2x - 1)$$

Q066 Written

Factorise $3x^3 + x^2 - 8x + 4$.**Solution**

1. Test divisor 1 and divide
2. *Factorquotient* $3x^2 + 4x - 4$
3. Therefore the final answer is $(x - 1)(x + 2)(3x - 2)$.

Final answer

$$(x - 1)(x + 2)(3x - 2)$$

Worked Solutions

Polynomial Theorems

Q067 Written

Factorise $2x^3 + 7x^2 + 8x + 3$.**Solution**

1. Test divisor -1 and divide twice
2. Completed factorisation
3. Therefore the final answer is $(x + 1)^2(2x + 3)$.

Final answer

$$(x + 1)^2(2x + 3)$$

Q068 MCQ

Factorise $x^3 + 2x^2 - x - 2$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 1)(x + 1)(x + 2)$.

Correct option: D**Final answer**

$$(x - 1)(x + 1)(x + 2)$$

Worked Solutions

Polynomial Theorems

Q069 MCQ

Factorise $x^3 - 2x^2 - 5x + 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 1)(x - 3)(x + 2)$.

Correct option: D**Final answer**

$$(x - 1)(x - 3)(x + 2)$$

Q070 MCQ

Factorise $x^3 - x^2 - 16x - 20$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x + 2)^2(x - 5)$.

Correct option: C**Final answer**

$$(x + 2)^2(x - 5)$$

Worked Solutions

Polynomial Theorems

Q071 MCQ

Factorise $2x^3 - x^2 - 5x - 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 2)(2x + 1)(x + 1)$.

Correct option: B**Final answer**

$$(x - 2)(2x + 1)(x + 1)$$

Q072 MCQ

Factorise $x^3 - 7x - 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 3)(x + 1)(x + 2)$.

Correct option: B**Final answer**

$$(x - 3)(x + 1)(x + 2)$$

Worked Solutions

Polynomial Theorems

Q073 MCQ

Factorise $x^3 + 4x^2 + x - 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 1)(x + 2)(x + 3)$.

Correct option: C**Final answer**

$$(x - 1)(x + 2)(x + 3)$$

Q074 MCQ

Factorise $x^3 + 6x^2 + 11x + 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x + 1)(x + 2)(x + 3)$.

Correct option: B**Final answer**

$$(x + 1)(x + 2)(x + 3)$$

Worked Solutions

Polynomial Theorems

Q075 MCQ

Factorise $x^3 - 3x + 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 1)^2(x + 2)$.

Correct option: D**Final answer**

$$(x - 1)^2(x + 2)$$

Q076 MCQ

Factorise $x^3 + x^2 - 8x - 12$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 3)(x + 2)^2$.

Correct option: D**Final answer**

$$(x - 3)(x + 2)^2$$

Worked Solutions

Polynomial Theorems

Q077 MCQ

Factorise $x^3 - 3x^2 - 9x - 5$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 5)(x + 1)^2$.

Correct option: D**Final answer**

$$(x - 5)(x + 1)^2$$

Q078 MCQ

Factorise $2x^3 - 3x^2 - 3x + 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 2)(x + 1)(2x - 1)$.

Correct option: A**Final answer**

$$(x - 2)(x + 1)(2x - 1)$$

Worked Solutions

Polynomial Theorems

Q079 MCQ

Factorise $3x^3 + x^2 - 8x + 4$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 1)(x + 2)(3x - 2)$.

Correct option: C**Final answer**

$$(x - 1)(x + 2)(3x - 2)$$

Q080 MCQ

Factorise $2x^3 + 7x^2 + 8x + 3$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x + 1)^2(2x + 3)$.

Correct option: D**Final answer**

$$(x + 1)^2(2x + 3)$$

Worked Solutions

Polynomial Theorems

Q081 Written

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$.**Solution**

1. $P(2) = 0$; divide by $x - 2$ twice
2. Remaining quadratic $x^2 - 2x - 4$
3. Therefore the final answer is $(x - 2)^2(x^2 - 2x - 4)$.

Final answer

$$(x - 2)^2(x^2 - 2x - 4)$$

Q082 Written

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$.**Solution**

1. Test roots and divide successively
2. Four linear factors
3. Therefore the final answer is $(x - 3)(x - 2)(x + 1)(x + 7)$.

Final answer

$$(x - 3)(x - 2)(x + 1)(x + 7)$$

Worked Solutions

Polynomial Theorems

Q083 Written

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$.**Solution**

1. $P(1) = 0$; *dividetwice*
2. *Remainingquadratic* $x^2 - 2x + 3$
3. Therefore the final answer is $(x - 1)^2(x^2 - 2x + 3)$.

Final answer

$$(x - 1)^2(x^2 - 2x + 3)$$

Q084 Written

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$.**Solution**

1. Test roots 1,2,3,4 successively
2. Four linear factors
3. Therefore the final answer is $(x - 4)(x - 3)(x - 2)(x - 1)$.

Final answer

$$(x - 4)(x - 3)(x - 2)(x - 1)$$

Worked Solutions

Polynomial Theorems

Q085 Written

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$.**Solution**

1. Test roots 1 and -2 successively
2. Completed factorisation
3. Therefore the final answer is $(x - 5)(x - 1)^2(x + 2)$.

Final answer

$$(x - 5)(x - 1)^2(x + 2)$$

Q086 Written

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$.**Solution**

1. Test roots 2, -2, 4, -3/2
2. Completed factorisation
3. Therefore the final answer is $(x - 4)(x - 2)(x + 2)(2x + 3)$.

Final answer

$$(x - 4)(x - 2)(x + 2)(2x + 3)$$

Worked Solutions

Polynomial Theorems

Q087 **Written****Factorise** $x^4 - 4x^3 - 4x^2 + 13x - 6$.**Solution**

1. Test roots 1,-2
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 1)(x + 2)(x^2 - 5x + 3)$.

Final answer

$$(x - 1)(x + 2)(x^2 - 5x + 3)$$

Q088 **Written****Factorise** $x^4 - 4x^3 + 3x^2 + 2x - 6$.**Solution**

1. Test roots 3,-1
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 3)(x + 1)(x^2 - 2x + 2)$.

Final answer

$$(x - 3)(x + 1)(x^2 - 2x + 2)$$

Worked Solutions

Polynomial Theorems

Q089 Written

Factorise $x^4 - x^3 + x^2 + x - 2$.**Solution**

1. Test roots 1,-1
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 1)(x + 1)(x^2 - x + 2)$.

Final answer

$$(x - 1)(x + 1)(x^2 - x + 2)$$

Q090 MCQ

Factorise $x^4 - 6x^3 + 8x^2 + 8x - 16$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 2)^2(x^2 - 2x - 4)$.

Correct option: D**Final answer**

$$(x - 2)^2(x^2 - 2x - 4)$$

Worked Solutions

Polynomial Theorems

Q091 MCQ

Factorise $x^4 + 3x^3 - 27x^2 + 13x + 42$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 3)(x - 2)(x + 1)(x + 7)$.

Correct option: A**Final answer**

$$(x - 3)(x - 2)(x + 1)(x + 7)$$

Q092 MCQ

Factorise $x^4 - 4x^3 + 8x^2 - 8x + 3$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 1)^2(x^2 - 2x + 3)$.

Correct option: D**Final answer**

$$(x - 1)^2(x^2 - 2x + 3)$$

Worked Solutions

Polynomial Theorems

Q093 MCQ

Factorise $x^4 - 10x^3 + 35x^2 - 50x + 24$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 4)(x - 3)(x - 2)(x - 1)$.

Correct option: D**Final answer**

$$(x - 4)(x - 3)(x - 2)(x - 1)$$

Q094 MCQ

Factorise $x^4 - 5x^3 - 3x^2 + 17x - 10$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 5)(x - 1)^2(x + 2)$.

Correct option: C**Final answer**

$$(x - 5)(x - 1)^2(x + 2)$$

Worked Solutions

Polynomial Theorems

Q095 MCQ

Factorise $2x^4 - 5x^3 - 20x^2 + 20x + 48$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 4)(x - 2)(x + 2)(2x + 3)$.

Correct option: D**Final answer**

$$(x - 4)(x - 2)(x + 2)(2x + 3)$$

Q096 MCQ

Factorise $x^4 - 4x^3 - 4x^2 + 13x - 6$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 1)(x + 2)(x^2 - 5x + 3)$.

Correct option: A**Final answer**

$$(x - 1)(x + 2)(x^2 - 5x + 3)$$

Worked Solutions

Polynomial Theorems

Q097 MCQ

Factorise $x^4 - 4x^3 + 3x^2 + 2x - 6$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 3)(x + 1)(x^2 - 2x + 2)$.

Correct option: B**Final answer**

$$(x - 3)(x + 1)(x^2 - 2x + 2)$$

Q098 MCQ

Factorise $x^4 - x^3 + x^2 + x - 2$ **Select the correct answer:****Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 1)(x + 1)(x^2 - x + 2)$.

Correct option: C**Final answer**

$$(x - 1)(x + 1)(x^2 - x + 2)$$

Worked Solutions

Polynomial Theorems

Q099 Written

Factorise $2x^3 - 3x^2 - x - 2$.**Solution**

1. $P(2) = 0$; quotient $2x^2 + x + 1$ has discriminant -7
2. Therefore the final answer is $(x - 2)(2x^2 + x + 1)$.

Final answer

$$(x - 2)(2x^2 + x + 1)$$

Q100 Written

Factorise $2x^3 - 7x^2 + 3x + 6$.**Solution**

1. Test divisor 2 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 2)(2x^2 - 3x - 3)$.

Final answer

$$(x - 2)(2x^2 - 3x - 3)$$

Worked Solutions

Polynomial Theorems

Q101 Written

Factorise $x^3 + 4x^2 + 2x - 4$.**Solution**

1. Test divisor -2 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x + 2)(x^2 + 2x - 2)$.

Final answer

$$(x + 2)(x^2 + 2x - 2)$$

Q102 Written

Factorise $x^3 - 3x^2 + 2$.**Solution**

1. Test divisor 1 and divide
2. Quadratic quotient remains
3. Therefore the final answer is $(x - 1)(x^2 - 2x - 2)$.

Final answer

$$(x - 1)(x^2 - 2x - 2)$$

Worked Solutions

Polynomial Theorems

Q103 MCQ

Factorise $2x^3 - 3x^2 - x - 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 2)(2x^2 + x + 1)$.

Correct option: C**Final answer**

$$(x - 2)(2x^2 + x + 1)$$

Q104 MCQ

Factorise $2x^3 - 7x^2 + 3x + 6$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 2)(2x^2 - 3x - 3)$.

Correct option: B**Final answer**

$$(x - 2)(2x^2 - 3x - 3)$$

Worked Solutions

Polynomial Theorems

Q105 MCQ

Factorise $x^3 + 4x^2 + 2x - 4$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x + 2)(x^2 + 2x - 2)$.

Correct option: B**Final answer**

$$(x + 2)(x^2 + 2x - 2)$$

Q106 MCQ

Factorise $x^3 - 3x^2 + 2$ Select the correct answer:**Solution**

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $(x - 1)(x^2 - 2x - 2)$.

Correct option: D**Final answer**

$$(x - 1)(x^2 - 2x - 2)$$

Worked Solutions

Polynomial Theorems

Q107 Challenge

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$.

Solution

1. *Substitute* $x = -1$
2. $(-1)^{57} + 57 = 56$
3. Therefore the final answer is 56.

Final answer

56

Worked Solutions

Polynomial Theorems

Q108 Challenge

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a , then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial.

Solution

1. Use the divisor zeros: $P(-1/2) = 0$ and $P(-2) = 0$.
2. Hence $a = 10$ for $2x + 1$, and $a = -11$ for $x + 2$.
3. Each linear factor corresponds to a polynomial root.

Final answer

$$a = 10 \quad \text{for } 2x + 1$$

$$a = -11 \quad \text{for } x + 2$$

linear factors $\leftrightarrow$ roots

Worked Solutions

Polynomial Theorems

Q109 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$.

Solution

1. $a + b = 0$ and $-3a + b = -4$
2. $R = x - 1$
3. Therefore the final answer is $x - 1$.

Final answer $x - 1$

Worked Solutions

Polynomial Theorems

Q110 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts.

Solution

1. $P = (2x - 1)^2(x^2 - 3)$
2. Remaining roots are $+ / - \sqrt{3}$
3. Therefore the final answer is $x = \sqrt{3}, -\sqrt{3}$.

Final answer

$$x = \sqrt{3}, -\sqrt{3}$$

Worked Solutions

Polynomial Theorems

Q111 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b .

Solution

1. $P(1) = 0$ gives $a + b = -10$
2. $P(-2) = 0$ gives $2a - b = -26$
3. $a = -12, b = 2$
4. Therefore the final answer is $a = -12, b = 2$.

Final answer

$$a = -12, b = 2$$

Worked Solutions

Polynomial Theorems

Q112 Challenge

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$.

Solution

1. $P = (x - 1)(x + 2)(x^2 - 4x - 6)$
2. Solve quadratic : $x = 2 \pm \sqrt{10}$
3. All four roots satisfy $P = 0$
4. Therefore the final answer is $x = 1, -2, 2 \pm \sqrt{10}$.

Final answer

$$x = 1, -2, 2 \pm \sqrt{10}$$

Worked Solutions

Polynomial Theorems

Q113 Challenge

Evaluate the exact remainder when $x^{57} + 57$ is divided by $x + 1$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 56.

Correct option: B

Final answer

56

Worked Solutions

Polynomial Theorems

Q114 Challenge

A model is $P(x) = 2x^3 - 3x^2 + ax + 6$. It must be divisible by $2x + 1$. Find a , then state what changes for divisor $x + 2$ and when more than one linear factor can divide a polynomial. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.

$$a = 10 \quad \text{for } 2x + 1$$

2. The correct value is $a = -11$ for $x + 2$.

linear factors $\leftrightarrow$ roots

Correct option: D

Final answer

$$a = 10 \quad \text{for } 2x + 1$$

$$a = -11 \quad \text{for } x + 2$$

linear factors $\leftrightarrow$ roots

Worked Solutions

Polynomial Theorems

Q115 Challenge

A polynomial $P(x)$ is divisible by $x - 1$ and leaves remainder -4 on division by $x + 3$. Find the remainder on division by $x^2 + 2x - 3$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $x - 1$.

Correct option: C

Final answer

$x - 1$

Worked Solutions

Polynomial Theorems

Q116 Challenge

The polynomial $P(x) = 4x^4 - 4x^3 - 11x^2 + 12x - 3$ has a horizontal intercept at $x = 1/2$ with multiplicity 2. Find the other intercepts. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $x = \sqrt{3}, -\sqrt{3}$.

Correct option: A

Final answer

$$x = \sqrt{3}, -\sqrt{3}$$

Worked Solutions

Polynomial Theorems

Q117 Challenge

Let $P(x) = x^4 - 3x^3 + ax^2 + bx + 12$. If both $x - 1$ and $x + 2$ are factors, find a, b Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $a = -12, b = 2$.

Correct option: C

Final answer

$$a = -12, b = 2$$

Worked Solutions

Polynomial Theorems

Q118 Challenge

For the polynomial above with $a = -12, b = 2$, determine all roots of $P(x) = 0$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $x = 1, -2, 2 \pm \sqrt{10}$.

Correct option: A

Final answer

$$x = 1, -2, 2 \pm \sqrt{10}$$

Worked Solutions

Polynomial Theorems

Quiz MCQ 1 [Concept Quiz · MCQ](#)

Find the remainder when $P(x) = x^3 - 2x + 5$ is divided by $x - 2$ Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is 9.

Correct option: C

Final answer

9

Quiz MCQ 2 [Concept Quiz · MCQ](#)

If $P(x)$ is divisible by $x + 1$, which statement is true? Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $P(-1) = 0$.

Correct option: C

Final answer

$P(-1) = 0$

Worked Solutions

Polynomial Theorems

Quiz MCQ 3 [Concept Quiz · MCQ](#)

The remainder on division by a quadratic divisor has the form $ax + b$. Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is $ax + b$.

Correct option: B

Final answer

$$ax + b$$

Quiz MCQ 4 [Concept Quiz · MCQ](#)

Which method records quotient and remainder for division by $x - a$? Select the correct answer:

Solution

1. Apply the same polynomial theorem as the source demand.
2. The correct value is *Syntheticdivision*.

Correct option: C

Final answer

Syntheticdivision

Worked Solutions

Polynomial Theorems

Quiz WRITTEN 5 Concept Quiz · Written

Find the remainder when $2x^3 + 3x^2 - x + 4$ is divided by $2x + 1$.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: 4

Final answer

4

Quiz WRITTEN 6 Concept Quiz · Written

Find the remainder $R = ax + b$ when $P(x)$ has values $P(1) = 3$ and $P(-2) = -6$.

Solution

1. Apply the relevant polynomial theorem.
2. Final answer: $R = 3x$

Final answer

$R = 3x$

Worked Solutions

Polynomial Theorems

Quiz WRITTEN 7 Concept Quiz · Written**Factorise** $x^3 - 6x^2 + 11x - 6$.**Solution**

1. Apply the relevant polynomial theorem.
2. Final answer: $(x - 1)(x - 2)(x - 3)$

Final answer

$$(x - 1)(x - 2)(x - 3)$$

Quiz WRITTEN 8 Concept Quiz · Written**If** $x - 1$ **and** $x + 2$ **are factors of** $x^4 - 3x^3 + ax^2 + bx + 12$, **find** a, b .**Solution**

1. Apply the relevant polynomial theorem.
2. Final answer: $a = -12, b = 2$

Final answer

$$a = -12, b = 2$$

Answer key

Use the question number shown on each page.

Q001 2	Q027 A	Q053 $Q = 3x^3 - x^2 + 2x - 6, R = 0$	Q079 C
Q002 6	Q028 A	Q054 D	Q080 D
Q003 C	Q029 A	Q055 $(x - 1)(x + 1)(x + 2)$	Q081 $(x - 2)^2(x^2 - 2x - 4)$
Q004 C	Q030 A	Q056 $(x - 1)(x - 3)(x + 2)$	Q082 $(x - 3)(x - 2)(x + 1)(x + 7)$
Q005 8	Q031 $3x - 2$	Q057 $(x + 2)^2(x - 5)$	Q083 $(x - 1)^2(x^2 - 2x + 3)$
Q006 $-\frac{1}{8}$	Q032 x	Q058 $(x - 2)(2x + 1)(x + 1)$	Q084 $(x - 4)(x - 3)(x - 2)(x - 1)$
Q007 0	Q033 $3x - 1$	Q059 $(x - 3)(x + 1)(x + 2)$	Q085 $(x - 5)(x - 1)^2(x + 2)$
Q008 4	Q034 $7x - 12$	Q060 $(x - 1)(x + 2)(x + 3)$	Q086 $(x - 4)(x - 2)(x + 2)(2x + 3)$
Q009 D	Q035 B	Q061 $(x + 1)(x + 2)(x + 3)$	Q087 $(x - 1)(x + 2)(x^2 - 5x + 3)$
Q010 A	Q036 B	Q062 $(x - 1)^2(x + 2)$	Q088 $(x - 3)(x + 1)(x^2 - 2x + 2)$
Q011 D	Q037 A	Q063 $(x - 3)(x + 2)^2$	Q089 $(x - 1)(x + 1)(x^2 - x + 2)$
Q012 A	Q038 D	Q064 $(x - 5)(x + 1)^2$	Q090 D
Q013 $a = 13/3$	Q039 $x^2 + 3x - 4$	Q065 $(x - 2)(x + 1)(2x - 1)$	Q091 A
Q014 $a = -4$	Q040 A	Q066 $(x - 1)(x + 2)(3x - 2)$	Q092 D
Q015 $a = -4$	Q041 $Q = x^2 - 2x + 9, R = -7$	Q067 $(x + 1)^2(2x + 3)$	Q093 D
Q016 $a = -16$	Q042 A	Q068 D	Q094 C
Q017 B	Q043 $Q = x^2 + x + 6, R = 0$	Q069 D	Q095 D
Q018 B	Q044 $Q = 3x^2 + 4x + 9, R = 26$	Q070 C	Q096 A
Q019 C	Q045 $Q = x^2 + 2x - 1, R = 4$	Q071 B	Q097 B
Q020 A	Q046 $Q = x^2 + 3x + 2, R = -1$	Q072 B	Q098 C
Q021 $a = -4$	Q047 $Q = 4x + 2, R = 4$	Q073 C	Q099 $(x - 2)(2x^2 + x + 1)$
Q022 A	Q048 B	Q074 B	Q100 $(x - 2)(2x^2 - 3x - 3)$
Q023 $2x + 1$	Q049 D	Q075 D	Q101 $(x + 2)(x^2 + 2x - 2)$
Q024 $-10x + 13$	Q050 D	Q076 D	Q102 $(x - 1)(x^2 - 2x - 2)$
Q025 $2x + 3$	Q051 A	Q077 D	Q103 C
Q026 $-15x + 19$	Q052 B	Q078 A	Q104 B

Q105 B**Q106** D**Q107** 56

$$a = 10 \quad \text{for } 2x + 1$$

Q108 $a = -11$ for $x + 2$ linear factors $\leftrightarrow$ roots**Q109** $x - 1$

Q110 $x = \sqrt{3}, -\sqrt{3}$

Q111 $a = -12, b = 2$

Q112 $x = 1, -2, 2 \pm \sqrt{10}$

Q113 B**Q114** D**Q115** C**Q116** A**Q117** C**Q118** A**Quiz MCQ 1** C**Quiz MCQ 2** C**Quiz MCQ 3** B**Quiz MCQ 4** C**Quiz WRITTEN 5** 4**Quiz WRITTEN 6** $R = 3x$ **Quiz WRITTEN 7** $(x - 1)(x - 2)(x - 3)$ **Quiz WRITTEN 8** $a = -12, b = 2$